

PC2174A: Mathematical Methods in Physics I

Assignment 7

1. [30 pts]

Two independent variables u, w are given in terms of two other independent variables x, y , by

$$u + w = x^2 + y^2 - k^2, \quad uw = a^2x^2 + b^2y^2 - h^4,$$

where a, b, h and k are constants. Show that

$$\begin{aligned} \left(\frac{\partial u}{\partial x}\right)_y &= -2x \left(\frac{a^2 - u}{u - w}\right), & \left(\frac{\partial w}{\partial y}\right)_x &= 2y \left(\frac{b^2 - w}{u - w}\right), \\ \left(\frac{\partial x}{\partial u}\right)_w &= -\frac{1}{2x} \left(\frac{b^2 - w}{a^2 - b^2}\right), & \left(\frac{\partial y}{\partial w}\right)_u &= \frac{1}{2y} \left(\frac{a^2 - u}{a^2 - b^2}\right). \end{aligned}$$

Answer:

Find the total differential of two given equations:

$$\begin{cases} du + dw = 2x \, dx + 2y \, dy \\ u \, dw + w \, du = 2a^2x \, dx + 2b^2y \, dy \end{cases} \quad (1)$$

(i) Let $dy = 0$, and we can derive:

$$dw = \frac{2a^2}{u}x \, dx - \frac{w}{u} \, du \quad (2)$$

As a result,

$$\left(1 - \frac{w}{u}\right) du = \left(2 - \frac{2a^2}{u}\right)x \, dx \Rightarrow \left(\frac{\partial u}{\partial x}\right)_y = -2x \left(\frac{a^2 - u}{u - w}\right) \quad (3)$$

(ii) It is similar to (i) since it is symmetric between $a \rightarrow b$ and $u \rightarrow w$, as a result:

$$\left(\frac{\partial w}{\partial y}\right)_x = -2y \left(\frac{b^2 - w}{w - u}\right) = 2y \left(\frac{b^2 - w}{u - w}\right) \quad (4)$$

(iii) Let $dw = 0$, and we can derive:

$$dy = \frac{1}{2y} \, du - \frac{x}{y} \, dx \quad (5)$$

As a result,

$$(w - b^2) \, du = (2a^2 - 2b^2) \, dx \Rightarrow \left(\frac{\partial x}{\partial u}\right)_w = -\frac{1}{2x} \left(\frac{b^2 - w}{a^2 - b^2}\right) \quad (6)$$

(iv) similar to (iii) because of the symmetric between $a \rightarrow b$ and $u \rightarrow w$

$$\left(\frac{\partial y}{\partial w}\right)_u = -\frac{1}{2y} \left(\frac{a^2 - u}{b^2 - a^2}\right) = \frac{1}{2y} \left(\frac{a^2 - u}{a^2 - b^2}\right) \quad (7)$$

2. [30 pts]

In quantum mechanics, the angular momentum operator $\hat{\mathbf{L}}$ is defined as

$$\hat{\mathbf{L}} = \sum_{k=1}^3 \hat{e}_k \hat{L}_k, \quad \hat{L}_k = -i\hbar \sum_{m=1}^3 \sum_{n=1}^3 \epsilon_{mnk} x_m \frac{\partial}{\partial x_n},$$

where $\hbar = h/2\pi$ is the reduced Planck constant. The raising/lowering operators are defined as

$$\hat{L}_{\pm} \equiv \hat{L}_x \pm i\hat{L}_y,$$

where $\hat{L}_x$ and $\hat{L}_y$ are x - and y -components of the angular momentum operator respectively. Transform the z -component of the angular momentum operator, $\hat{L}_z$, and the raising operator, $\hat{L}_+$, into the spherical coordinates (r, θ, ϕ) given by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

Answer:

It's trivial to find the relations:

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \rightarrow dr = \frac{x}{r} dx + \frac{y}{r} dy + \frac{z}{r} dz \\ \tan \phi = \frac{y}{x} \rightarrow \frac{d\phi}{\cos^2 \phi} = -\frac{y}{x^2} dx + \frac{dy}{x} \\ \tan \theta = \frac{\sqrt{x^2 + y^2}}{z} \rightarrow \frac{d\theta}{\cos^2 \theta} = \frac{x}{z\sqrt{x^2 + y^2}} dx + \frac{y}{z\sqrt{x^2 + y^2}} dy - \frac{\sqrt{x^2 + y^2}}{z^2} dz \end{cases} \quad (8)$$

The transformation between these two coordinate differentials is thus derived

$$\begin{aligned} \frac{\partial r}{\partial x} &= \sin \theta \cos \phi, \quad \frac{\partial r}{\partial y} = \sin \theta \sin \phi, \quad \frac{\partial r}{\partial z} = \cos \theta \\ \frac{\partial \phi}{\partial x} &= -\frac{y \cos^2 \phi}{x^2} = -\frac{r \sin \theta \sin \phi \cos^2 \phi}{r^2 \sin^2 \theta \cos^2 \phi} = -\frac{\sin \phi}{r \sin \theta} \\ \frac{\partial \phi}{\partial y} &= \frac{\cos^2 \phi}{x} = \frac{\cos^2 \phi}{r \sin \theta \cos \phi} = \frac{\cos \phi}{r \sin \theta}, \quad \frac{\partial \phi}{\partial z} = 0 \\ \frac{\partial \theta}{\partial x} &= \frac{x \cos^2 \theta}{z \sqrt{x^2 + y^2}} = \frac{r \sin \theta \cos \phi}{r^2 \cos \theta \sin \theta} = \frac{1}{r} \cos \phi \cos \theta \\ \frac{\partial \theta}{\partial y} &= \frac{y \cos^2 \theta}{z \sqrt{x^2 + y^2}} = \frac{r \sin \theta \sin \phi}{r^2 \cos \theta \sin \theta} = \frac{1}{r} \sin \phi \cos \theta \\ \frac{\partial \theta}{\partial z} &= -\frac{\sqrt{x^2 + y^2} \cos^2 \theta}{z^2} = -\frac{r \sin \theta \cos^2 \theta}{r^2 \cos^2 \theta} = -\frac{\sin \theta}{r} \end{aligned}$$

Express $\frac{\partial}{\partial x_n}$ in spherical coordinates:

$$\begin{aligned} \frac{\partial}{\partial x} &= \sin \theta \cos \phi \frac{\partial}{\partial r} - \frac{\sin \phi}{r \sin \theta} \frac{\partial}{\partial \phi} + \frac{1}{r} \cos \phi \cos \theta \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial y} &= \sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{\cos \phi}{r \sin \theta} \frac{\partial}{\partial \phi} + \frac{1}{r} \sin \phi \cos \theta \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} &= \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \end{aligned} \quad (9)$$

Substitute the above relation into the components of the angular momentum operator,

$$\hat{L}_x = -i\hbar(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}) = i\hbar(\sin \phi \frac{\partial}{\partial \theta} + \cos \phi \cot \theta \frac{\partial}{\partial \phi}) \quad (10)$$

$$\hat{L}_y = -i\hbar(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}) = i\hbar(-\cos \phi \frac{\partial}{\partial \theta} + \sin \phi \cot \theta \frac{\partial}{\partial \phi}) \quad (11)$$

$$\hat{L}_z = -i\hbar(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}) = -i\hbar \frac{\partial}{\partial \phi} \quad (12)$$

And the raising operator is

$$\begin{aligned} \hat{L}_+ &= \hat{L}_x + i\hat{L}_y = \hbar \left[(\cos \phi + i \sin \phi) \frac{\partial}{\partial \theta} + i(\cos \phi + i \sin \phi) \cot \theta \frac{\partial}{\partial \phi} \right] \\ &= \hbar e^{i\phi} \left[\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right] \end{aligned} \quad (13)$$

3. [40 pts]

In analytical mechanics, the central mathematical entity is the Lagrangian $L(q(t), \dot{q}(t), t)$, which is treated as a function of generalized coordinates q_i , generalized velocities $\dot{q}_i \equiv \frac{dq_i}{dt}$ for $i = 1, 2, \dots, N$, and time t .

- (a) The conjugate momentum p_i associated to the generalized coordinate q_i is defined as $p_i \equiv \frac{\partial L}{\partial \dot{q}_i}$. In analytical mechanics, the Hamiltonian is defined as

$$H(q(t), p(t), t) \equiv \sum_{i=1}^N \dot{q}_i(t) p_i(t) - L(q(t), \dot{q}(t), t),$$

and the Euler–Lagrange equation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0.$$

Show that

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}, \quad i = 1, 2, \dots, N.$$

- (b) The Poisson bracket $[F, G]_{\text{PB}}$ of two arbitrary dynamical functions $F(q(t), p(t), t)$ and $G(q(t), p(t), t)$ is defined by

$$[F, G]_{\text{PB}} \equiv \sum_{i=1}^N \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right).$$

Prove the following fundamental Poisson brackets:

$$[q_i, q_j]_{\text{PB}} = 0, \quad [p_i, p_j]_{\text{PB}} = 0, \quad [q_i, p_j]_{\text{PB}} = \delta_{ij}, \quad i = 1, 2, \dots, N.$$

- (c) Show that the time derivative of an arbitrary dynamical function $F(q(t), p(t), t)$ is

$$\frac{dF}{dt} = [F, H]_{\text{PB}} + \frac{\partial F}{\partial t}.$$

Hence, deduce that any Hamiltonian which does not depend on time explicitly is a constant of motion.

Answer:

- (a) Considering q , p , t as independent variables, only L includes t . As a result,

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \quad (14)$$

And the other two relations can be derived easily through directly calculation:

$$\frac{\partial H}{\partial p_i} = \frac{\partial}{\partial p_i} \left(\sum_j \dot{q}_j p_j \right) - \frac{\partial L}{\partial p_i} = \sum_j \dot{q}_j \delta_{ij} = \dot{q}_i \quad (15)$$

$$\frac{\partial H}{\partial q_i} = \frac{\partial}{\partial q_i} \left(\sum_j \dot{q}_j p_j \right) - \frac{\partial L}{\partial q_i} = -\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = -\dot{p}_i \quad (16)$$

- (b) Since $\frac{\partial p_i}{\partial q_j} = \frac{\partial q_i}{\partial p_j} = 0$, the first two Poisson brackets equal to 0. And the third one:

$$[q_i, p_j]_{\text{PB}} = \sum_k \left(\frac{\partial q_i}{\partial q_k} \frac{\partial p_j}{\partial p_k} - \frac{\partial p_j}{\partial q_k} \frac{\partial q_i}{\partial p_k} \right) = \sum_k \delta_{ik} \delta_{jk} = \delta_{ij} \quad (17)$$

- (c)

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i \right) + \frac{\partial F}{\partial t} \quad (18)$$

Substitute the results of (a) into equation (18)

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right) + \frac{\partial F}{\partial t} = [F, H] + \frac{\partial F}{\partial t} \quad (19)$$

Hamiltonian which does not depend on time explicitly implies that

$$\frac{\partial H}{\partial t} = 0 \quad (20)$$

As a result.

$$\frac{dH}{dt} = [H, H] + \frac{\partial H}{\partial t} = 0 \quad (21)$$

Q.E.D.

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